

The hidden trends in battery supply and demand: A regional analysis

Global trends suggest that the battery market is oversaturated, but a regional analysis reveals pockets of opportunity.

This article is a collaborative effort by Jakob Fleischmann, Martin Linder, and Raphael Rettig, with Alexandre Van de Rijt and Christian Ahrberg, representing views from McKinsey's Automotive & Assembly Practice.



Although electric-vehicle (EV) sales have slowed from their peak, battery technology continues to evolve at a breakneck pace. Researchers are constantly experimenting with new chemistries and cell configurations to optimize battery range, charging speed, and vehicle cost—the factors that matter most to consumers. Many automotive OEMs, still convinced that the future is electric, have helped global battery demand rise. Simultaneously, many governments worldwide continue to offer incentives to encourage battery production as part of their efforts to combat climate change and give consumers greener transportation options.

Many battery suppliers are enthusiastic about their prospects and have been ramping up production. This trend, combined with lower-than-expected EV demand, has led to a global battery oversupply. Some analysts are concerned that the imbalance may persist, especially if battery incumbents maintain their high output and new start-ups continue to enter the mix.

Looking at the global picture, battery suppliers appear to be in a tough situation that could force them to reduce prices or cut their output. But a recent McKinsey analysis provides a more nuanced view of the market, showing that their prospects vary greatly by region. While some countries produce more batteries than they need, others rely on imports because domestic manufacturing cannot fulfill demand. The same pattern holds true for the upstream value chain, including raw materials.

Many industries can eliminate regional supply–demand imbalances through global trade, but the battery market's unique features, including greater regulatory limitations, trade barriers, high shipping costs, and variations in upstream-material availability, complicate this strategy. While imports now address shortages, they may not be sufficient to fill the gap if demand grows. In consequence, some countries may continue to have a battery surplus while others may still experience shortages—and that means companies along the value chain will need region-specific strategies to capture opportunities. These strategies could become even more important if [trade barriers](#) intensify, further limiting imports and exports.

The global view: An unbalanced market in which supply exceeds demand

Before we examined regional trends for batteries, we first reviewed the global market to understand the overall dynamics. Our analysis relied on a bottom-up model that reviewed projected global battery supply in combination with major demand drivers, such as [electric vehicles](#), energy storage applications, and consumer electronics. Note that all our analyses focus on batteries that rely on [lithium-type chemistries](#), such as lithium iron phosphate (LFP) and lithium nickel manganese cobalt mixed oxide (NMC).

Although OEMs have mainly used NMC batteries for many years, LFP and a new variant containing iron and magnesium, termed L(M)FP, have been gaining popularity because recent technological advances have increased their energy density and driving range at the pack level. L(M)FP now accounts for about half the market for cathode active material (CAM). (For more

information on battery chemistries, see sidebar “How battery technology is evolving.”) Our model estimates that lithium-ion batteries are expected to account for about 86 percent of total battery demand by 2030, with sodium-ion, lead-acid, and other battery technologies accounting for the remaining 14 percent.

Global demand

We created three global demand scenarios for batteries: fading momentum, continuation of the current trajectory (base case), and further acceleration. The main demand differentiators included variations in EV production volume and uptake of energy storage systems. In the base case scenario, worldwide demand is expected to rise from about 1,970 gigawatt-hours (GWh) in 2025 to around 3,910 GWh by 2030 (Exhibit 1).

Global supply

We estimated how battery suppliers in China, Europe, North America, and the rest of the world are planning to expand in response to high demand. If they fulfill their stated goals, their announced nameplate capacity would reach about 4,340 GWh by 2025 and 6,440 GWh by 2030.

Companies are unlikely to meet nameplate capacity because complications often arise during production. To determine actual output, we first adjusted capacity estimates based on the number of gigawatt-hours likely to be lost because of construction delays, ramp-up inefficiencies, and yield loss during the initial production years. Ramp-up inefficiencies had the greatest impact on capacity.

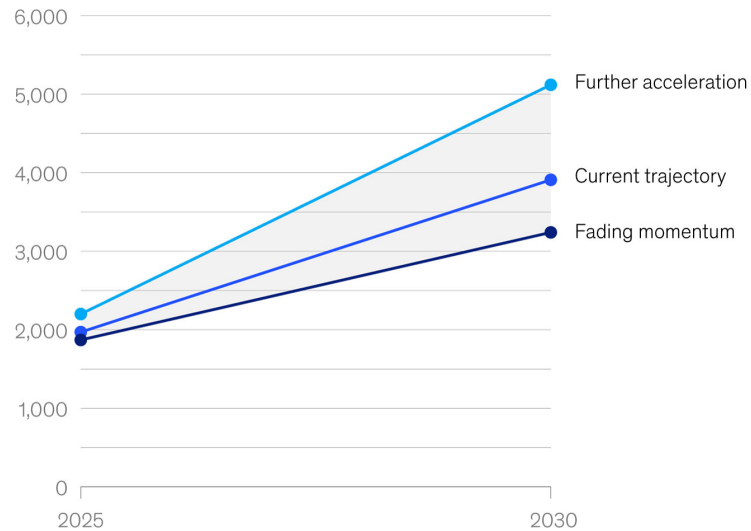
How battery technology is evolving

Batteries can use various cathode active materials (CAMs), including lithium-ion, lead acid, nickel-metal hydride, and sodium-ion chemistries. For the past ten years, automotive OEMs have usually defaulted to lithium nickel manganese cobalt mixed oxide (NMC) chemistry because it provided the longest driving range. Recently, however, lithium iron phosphate (LFP) and a new iron and manganese form, termed L(M)FP, have been attracting more attention because technological improvements have increased their driving range. The L(M)FP-type chemistries also cost less than NMC chemistry and have a better safety profile.

Exhibit 1

Global battery demand is expected to increase through 2030 in all scenarios.

Global battery demand in 2025 and 2030, by scenario, gigawatt-hours



Source: Consumer Electronics 2025 © 2025 Euromonitor International Limited, all rights reserved; Light Vehicle Production Forecast, S&P Global Mobility, Apr 4, 2025; McKinsey Battery Insights; McKinsey Energy Storage Insights; vehicle specifications per McKinsey analysis

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We then scored all companies based on their likelihood of reaching production goals. Our scoring system considered a range of factors, such as a company's experience, intellectual property, offtake agreements, and revenue streams. This system allowed us to create three supply scenarios that show the realistic production capacity for the following groups:

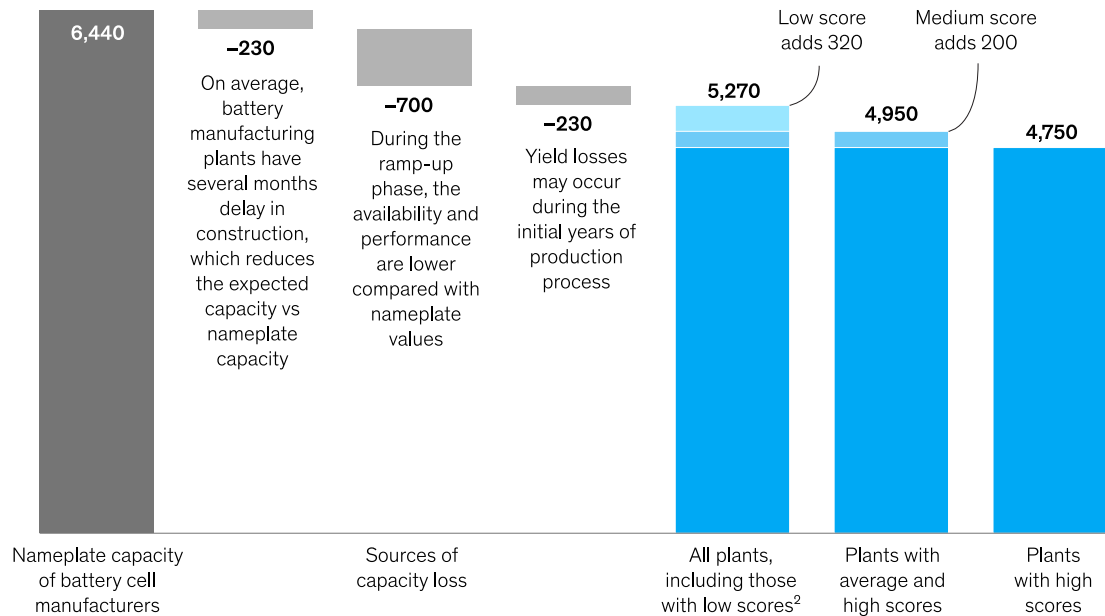
- all companies in our analysis
- only companies with an average score or higher (base case)
- only companies with a high score

Under this analysis, the 2030 supply varies from 4,750 GWh (only high-scoring companies) to 5,270 for all companies—both well below the 6,440 GWh announced nameplate capacity (Exhibit 2).

Exhibit 2

Projected battery supply in 2030 is expected to be below the announced nameplate capacity in all scenarios.

Potential global battery supply, by supplier score,¹ gigawatt-hours



¹A scoring system evaluates likelihood of realization based on a set of factors, including company experience, intellectual property, revenue streams, and offtake plans.

²All announcements, including plants that are operational, under construction, and in early and advanced planning stage.

Source: McKinsey Battery Insights

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Global supply–demand mismatch

We then considered our supply and demand scenarios in different combinations. If demand slows or remains on its current trajectory, there will be an oversupply of batteries in 2025 and 2030, even if only the highest-scoring companies meet their production goals. The oversupply gets even larger if the estimate is expanded to include all companies, or even just those with average rankings (Exhibit 3).

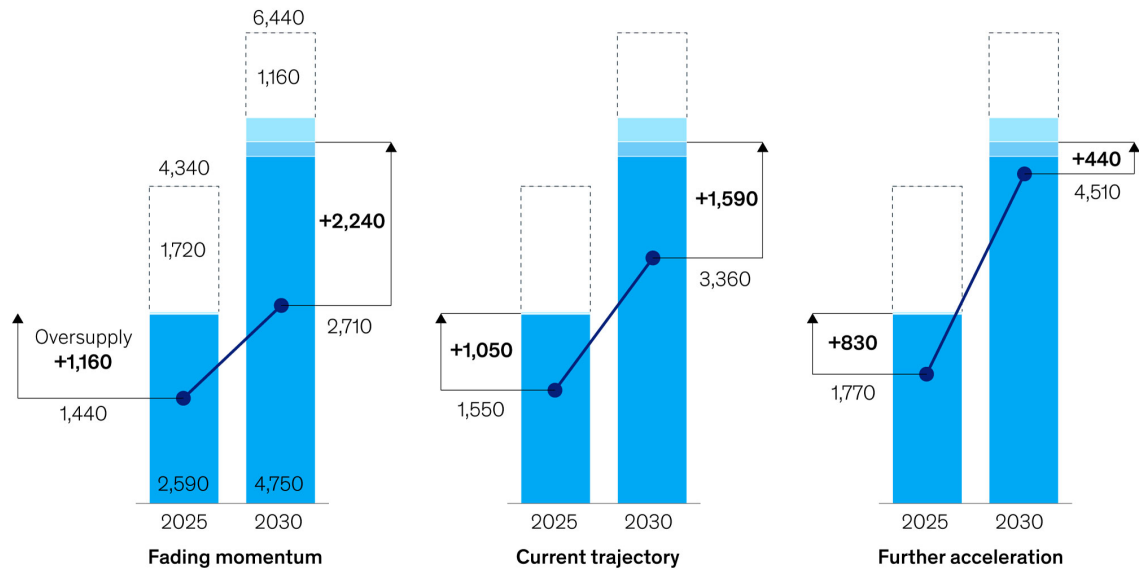
The further acceleration scenario also depicts a battery oversupply. If companies with a high or average score meet their production goals, there would be an oversupply of 830 GWh in 2025 and 440 GWh in 2030.

Exhibit 3

A global battery surplus is likely through 2030 under different supply–demand scenarios.

Lithium-ion battery demand and supply in 2025 and 2030, by scenario and company score,¹
gigawatt-hours

Demand ●—● **Supply** ■ High scoring ■ Medium scoring ■ Low scoring □ Total with announced nameplate supply²



¹A scoring system evaluates likelihood of realization based on a set of factors, including company experience, intellectual property, revenue streams, and offtake plans.

²All announcements, including plants that are operational, under construction, and in early and advanced planning stage.

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Regional revelations: A varied picture of supply and demand at the local level

Although battery supply may exceed demand at the global level, the picture is more nuanced and varied by region. Some countries have excess capacity—meaning more than enough to satisfy local demand—while others rely on imports to alleviate local shortages. This regional view could become critical if more countries try to localize production.

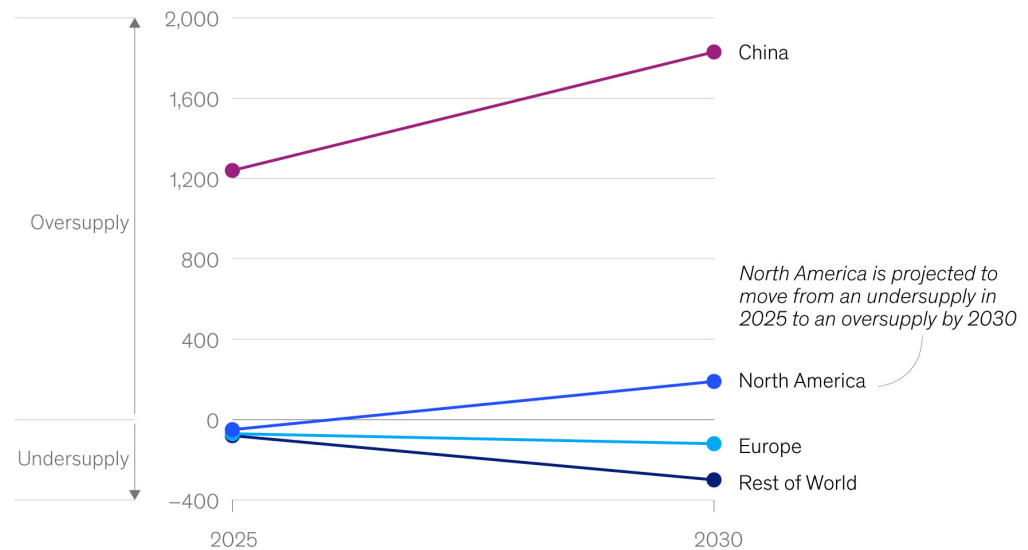
In our base case scenario, in which demand continues to grow according to the current trajectory, China's current oversupply of lithium-ion batteries—both L(M)FP and NMC—persists

through 2030 even if all announced capacity materializes (Exhibit 4). North America shifts from an undersupply in 2025, which is currently addressed through imports, to an oversupply in 2030. All other regions will not have enough local capacity to satisfy domestic demand.

Exhibit 4

Supply and demand for lithium-ion batteries will be unbalanced at the regional level in 2025 and 2030.

Lithium-ion battery over- or undersupply in 2025 and 2030, by current trajectory scenario, gigawatt-hours



Source: Consumer Electronics 2025 © 2025 Euromonitor International Limited, all rights reserved; Light Vehicle Production Forecast, S&P Global Mobility, Apr 4, 2025; McKinsey Battery Insights; McKinsey Energy Storage Insights; vehicle specifications per McKinsey analysis

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About the Battery Accelerator Team

McKinsey's Battery Accelerator Team consists of more than 350 experts across Asia, Europe, and North America. It includes senior industry experts with deep knowledge of cell R&D, gigafactory construction, and industrialization. The team provides tailored solutions to those active in the battery ecosystem, including companies focusing on materials, manufacturing, storage, and recycling.

When examining regional supply and demand for specific lithium-ion chemistries, the picture becomes even more nuanced. For years, Europe and North America have focused on producing NMC and nickel cobalt aluminum (NCA) batteries. They have yet to increase their capacity for L(M)FP-type chemistries, even though recent technology advances and geopolitical events have increased their appeal. China, which began to favor L(M)FP chemistries before other regions did, now has a significant oversupply, while other regions have a large undersupply and require L(M)FP battery imports. These trends are expected to continue through 2030 (Exhibit 5). For NMC and NCA batteries, all regions are expected to have an oversupply in 2030.

What factors contribute to these regional supply–demand imbalances? To answer this question, we examined current regional capacity in more detail and investigated the trends shaping local markets.

China

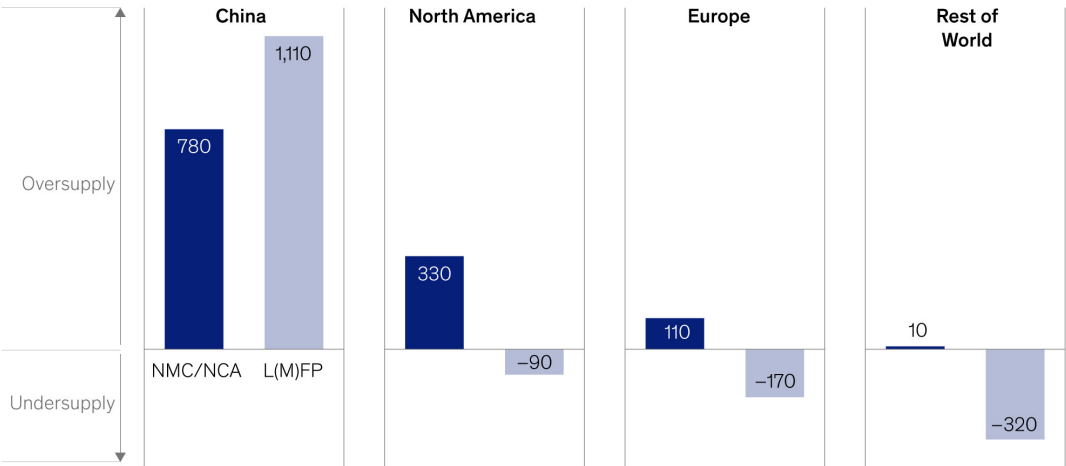
Globally, the top two battery suppliers are Chinese. They conduct almost all production within China, but many of their customers, including major OEMs, are headquartered elsewhere. About 64 percent of Chinese battery capacity in 2030 will involve L(M)FP-type chemistries.

Despite China’s high export volume, the local oversupply is so large that many battery manufacturing plants are not operating at capacity. Some Chinese companies have recently decreased L(M)FP prices to encourage utilization and become more attractive to customers, which has exacerbated margin pressures.

Although Chinese companies face many challenges, they have a strong supply of battery materials. For instance, they have 67 percent of global lithium-refining capacity (Exhibit 6). Other regions, by contrast, must optimize sourcing and capabilities related to upstream raw materials and [cell components](#) before they can localize more battery cell production.

Exhibit 5
In 2030, lithium-ion batteries are expected to be in short supply in all regions except China.

Supply over demand of NMC/NCA¹ batteries and L(M)FP² batteries by 2030, gigawatt-hours

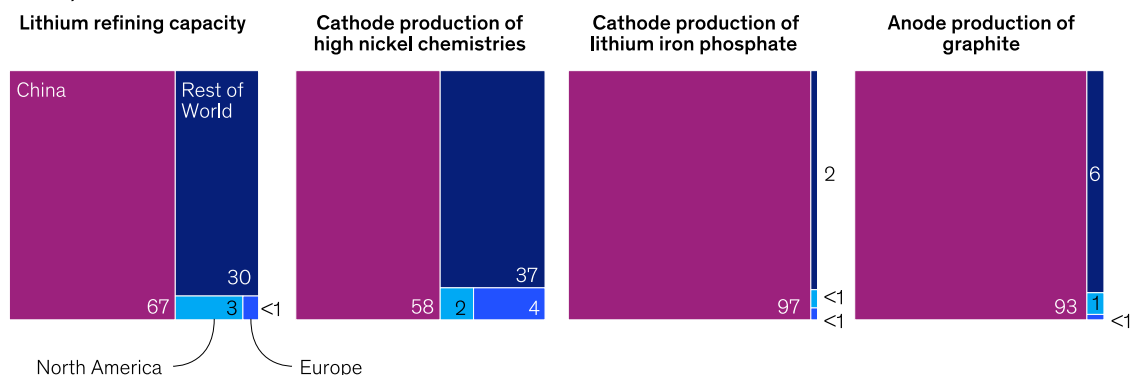


¹Lithium nickel manganese cobalt mixed oxide and lithium nickel cobalt aluminum.
²Lithium iron (manganese) phosphate.
Source: Consumer Electronics 2025 © 2025 Euromonitor International Limited, all rights reserved; Light Vehicle Production Forecast, S&P Global Mobility, Apr 4, 2025; McKinsey Battery Insights; McKinsey Energy Storage Insights; vehicle specifications per McKinsey analysis

Exhibit 6

Chinese companies have the largest market share for many critical battery materials.

Global market share for key battery materials, by installed adjusted production capacity and region, 2025, %



Note: Figures do not sum to 100%, because of rounding.
Source: McKinsey Battery Insights; McKinsey MineSpans

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Europe

Under our current trajectory scenario, Europe is expected to have a lithium-ion battery undersupply of around 70 GWh in 2025 even if all announced projects convert to production. The gap will need to be addressed through battery imports. It is likely that the deficit will be higher, however, since start-ups account for about 30 percent of expected battery production volume, which is significantly higher than in China or North America. Start-ups have a greater risk of delays in construction and ramp-up issues, partly because many are betting on untested technological innovations that could result in lower efficiency and yield.

If the lithium-ion battery undersupply persists through 2030, as expected, European cell suppliers could face more competition from Chinese imports that offer sophisticated technologies at attractive prices.

The European Union has recently approved some policies that may benefit the battery industry, including the Green Deal Industrial Plan, which is designed to promote net-zero sectors, and the EU Battery Regulation, which aims to increase sustainability. But these policies alone cannot counter all headwinds facing the industry, some of which have led many European battery companies to delay or cancel projects, or even declare bankruptcy, in recent years. The new EU Competitiveness Compass, which was designed to encourage innovation across industries, contains several plans that might provide some additional relief and increase local battery production. For instance, the Clean Industrial Deal and the Industrial Action Plan for the European Automotive Sector are intended to promote both green technologies and local manufacturing.

North America

North America is now expected to have a 50 GWh undersupply of lithium-ion batteries in 2025 under our current trajectory scenario, which will require imports from other regions to supplement domestic production. There is still a possibility that the market could become

balanced by 2030, however, because the US Inflation Reduction Act (IRA) includes subsidies for local manufacturing. (It also includes incentives for purchasing EVs, which could accelerate battery demand, meaning that even more capacity might be needed to balance the market.) The future of the IRA is unknown, however, since it was created during the previous administration, and the [potential impact of US tariffs adds to the uncertainty](#).

The upstream supply chain for active materials, electrolytes, separators, and other components is still relatively underdeveloped in North America, especially for L(M)FP-type chemistries. If local battery manufacturers have difficulty sourcing these materials, it could limit the extent to which they can ramp up production.

Rest of world

In the rest of the world (that is, the regions outside China, Europe, and North America), the current undersupply of lithium-ion cells could rise from 80 GWh in 2025 to 300 GWh by 2030. The major demand drivers for rest-of-world regions vary by country and include the following:

- all types of electric mobility in Japan and South Korea
- energy [storage systems](#) in Australia
- the expansion of two- and three-wheeled EVs in India, Indonesia, and Thailand

While Japan and South Korea already have local battery value chains capable of meeting domestic demand, other rest-of-world countries must import products from oversupplied regions. Some of these countries are introducing policies to spur local production or plan to do so. For instance, Indonesia announced purchase subsidies for EVs in March 2023, with the goal of encouraging domestic EV manufacturing and strengthening the local supply chain.

Mitigation strategies and opportunities for value creation

While the global market for batteries is facing a potential oversupply, significant opportunities exist by region and chemistry. After identifying the areas of greatest demand, companies can create targeted strategies to capture value, but these will vary greatly by company type and location.

Upstream players

About 30 percent of global L(M)FP CAM is imported to Europe and North America from China, but leaders in both the public and private sectors hope to become more self-sufficient by increasing the local supply. This goal will require substantial funding from both private and public investors to help upstream companies increase their capacity.

About the McKinsey Center for Future Mobility

These insights were developed by the [McKinsey Center for Future Mobility \(MCFM\)](#). Since 2011, MCFM has worked with stakeholders across the mobility ecosystem by providing independent and integrated evidence about possible future-mobility scenarios. With our unique, bottom-up modeling approach, our insights enable an end-to-end analytics journey through the future of mobility—from consumer needs to a modal mix across urban and rural areas, sales, value pools, and life cycle sustainability.

As European and North American companies expand, they can reduce risk through offtake agreements in which local buyers commit to purchasing a specific amount of CAM. In addition to helping upstream players gauge demand more accurately, these agreements guarantee a cash flow. Offtake agreements may be particularly beneficial to companies that lack traditional funding sources.

Production costs in Europe and North America are likely to be higher than those in China, but government incentives, such as those in the US Inflation Reduction Act, may provide some relief. What's more, cell manufacturers and OEMs may give preference to local upstream companies because this could potentially increase their eligibility for government-funded incentives.

Cell manufacturers

For cell manufacturers, strategies for capturing value will vary by region. In China, where companies have experienced years of significant growth, cell manufacturers may increasingly focus on managing costs and optimizing utilization. This strategy could potentially slow any local expansion plans. Since most cell-manufacturing costs—generally about 30 to 40 percent—relate to raw material inputs, Chinese manufacturers may investigate partnerships with upstream suppliers that provide benefits, such as discounts or the opportunity to reduce risk through coinvestments. They may also pursue new battery chemistries, such as moving from the traditional LFP to L(M)FP, to improve profitability. Such shifts might allow them to attain greater energy density while maintaining costs.

In other locations, cell manufacturers could strive to improve multiple dimensions simultaneously:

- managing costs to reduce the gap with Chinese companies
- ramping up or scaling local factories to gain domestic market share and relevance
- improving local sourcing of upstream materials to reduce supply chain risk and possibly benefit from government incentives
- continuously advancing technologies to remain competitive

While companies may capture many opportunities through a multipronged effort, this strategy also poses numerous challenges. Companies can increase their chances of success by setting clear priorities and being stringent about meeting goals. This approach will likely increase efficiency and profitability more quickly, which will help maintain support from investors and stakeholders.

Downstream users

Automotive EV OEMs are very concerned about supply chain stability and increasingly seek resilient upstream inputs, often by contracting with two or more suppliers. European and North American/Mexican companies are also attempting to increase the number of local companies in the end-to-end supply chain to minimize [geopolitical risks](#), shorten transit times, and increase the likelihood of qualifying for domestic government incentives.

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While many OEMs are already planning to source cells locally, either by manufacturing them internally or by forming joint ventures or partnerships with other companies, most do not have a strategy for localizing upstream inputs, including those related to refined or active materials.

Stationary storage companies in Europe and North America continue to face challenges in securing L(M)FP because most domestic cell capacity involves NMC. To ensure their future supply, they can follow one of two paths. The first involves locking in local volume commitments, while the second involves leveraging the global L(M)FP oversupply, which largely results from China's high level of production. Purchasing cells from China could also decrease overall costs, given the efficiency of Chinese companies. As with other components, offtake agreements with suppliers can help reduce risk and strengthen the supply chain.

At the global level, the battery industry appears to have a persistent oversupply. A regional analysis reveals a much more nuanced picture, however, with demand varying by both location and chemistry. These differences primarily result from a vast oversupply of L(M)FP-type cells in China and an undersupply within Europe and North America. A regional view of battery trends will become even more important as governments increasingly attempt to localize the battery supply chain and as consumer preferences, including attitudes toward EVs, continue to evolve. To capture regional opportunities, companies should act fast to ensure reliable and cost-effective sourcing, prepare for shifts in demand, and have a clear idea of regional supply–demand trends.

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